Locked Waveform output (like an oscilloscope) in VB.NET

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- Thursday, August 04, 2011 11:17 AM

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I'm making a PC oscilloscope with USB interface. I'm quite familiar with USB HID programming. The basic principle is simple. The microcontroller takes 50 samples (ie. buffers) and then send it as reports. Now the program acquires it and shows the waveform.

Now, the problem is how to display the waveform perfectly like a real oscilloscope. I tried using the Chart Control. I never used it before. I simply added:

```
With Me.Chart1.Series(0)

     If x = 51 Then
       .Points.Clear()
     End If

     .Points.AddXY(x, sample)
     .ChartType = DataVisualization.Charting.SeriesChartType.Spline

     .BorderWidth = 1
   End With
```

I know it's not the correct way to visualize the waveform. It simply plots the acquired samples (ranging from 0 to 1024) along y axis against x axis (ranging 0 to 50, that's why there's the .Pints.clear at x= 51). Of course, it doesn't look like a waveform.

Any idea how to do it?

It might be solved by creating a database, which will have two columns, 'Count' and 'Sample'. The buffered samples will be put into the 'sample' rows against 'count' from 0 to 50. And the database will be periodically update whenever a HID report containing the microcontroller ADC reading samples is received, usually at 500ms interval. But I can not code it. Because I don't have programming experiences using databases.

Any other idea? I might the possible to plot the lines manually (without using the chart control) using the graphics library.

What do you think?

All Replies

- Thursday, August 04, 2011 12:38 PM

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  Creating Line Graph in Vb.net

  http://social.msdn.microsoft.com/Forums/en-US/vbgeneral/thread/f59a7bec-dd47-4019-a703-b3ce5760181b

```
Dim hght, num As Byte
 Dim x As Integer

 Private Sub tmrsec_Tick(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles tmrsec.Tick
  num = Int(Rnd() * 180)

  lblval.Text = num & " " & hght

  lblperformance.CreateGraphics.DrawLine(New Pen(Color.Blue, 2), x, hght, x, num)
  If x <= 595 Then
   x += 4
  End If

  If x > 595 Then x = 0 : lblperformance.Refresh()
 End Sub
```

There are several examples in the link for creating graph, The code i pasted creates a graph by drawing straight line by random values of X and Y.

Use this second example to create a single line graph (wave form), uses random values of X and Y axis.

```
Public Class Form1
  Dim ht, num As Byte
  Dim x As Integer

  Private Sub Timer1_Tick(sender As System.Object, e As System.EventArgs) Handles Timer1.Tick
    num = Int(Rnd() * 180)

    lblval1.Text = num

    'lblperformance.CreateGraphics.DrawLine(New Pen(Color.Blue, 2), x, ht, x + 10, num)

    lblperformance.CreateGraphics.DrawLine(New Pen(Color.PaleGreen, 2), 0, 50, 600, 50)
    lbl50.Location = New Point(lblperformance.Left - 30, lblperformance.Top + 50)

    lblperformance.CreateGraphics.DrawLine(New Pen(Color.PaleGreen, 2), 0, 100, 600, 100)
    lbl100.Location = New Point(lblperformance.Left - 30, lblperformance.Top + 100)

    lblperformance.CreateGraphics.DrawLine(New Pen(Color.PaleGreen, 2), 0, 150, 600, 150)
    lbl150.Location = New Point(lblperformance.Left - 30, lblperformance.Top + 150)

    lblperformance.CreateGraphics.DrawLine(New Pen(Color.Blue, 2), x, ht, x + 10, num)

    ht = num

    If x <= 595 Then
      x += 10
    End If

    If x > 595 Then x = 0 : lblperformance.Refresh()
  End Sub
```

```
  Private Sub Form1_Load(sender As Object, e As System.EventArgs) Handles Me.Load
    x = 40
    ' ht = 0
  End Sub
End Class
```

Faraz

- Thursday, August 04, 2011 6:25 PM

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You should use a ChartType of .Line (or .FastLine).

Take a look at this simple-minded example:

```
  Private Sub Form1_Load(sender As System.Object, e As System.EventArgs) Handles MyBase.Load
    Timer1.Enabled = True
    Chart1.Series(0).ChartType = DataVisualization.Charting.SeriesChartType.FastLine
  End Sub

  Private Sub Timer1_Tick(sender As Object, e As System.EventArgs) Handles Timer1.Tick
    With Chart1.Series(0)
      .Points.Clear()
      For I = 0 To 99
        .Points.AddY(Rnd())
      Next
    End With
  End Sub
```

I set the Timer.Interval = 1000, so that I would generated a new plot for each 1-second tick. I used Rnd to generate a random waveform, so (naturally), your actual data will be from a "real" source, not a random number generator.

If you want faster response than you can use the Add method, intead of the AddY, where you furnish a populated array, intead of individual points.

Dick

Dick Grier. Author of Visual Basic Programmer's Guide to Serial Communications 4. See www.hardandsoftware.net.

- Friday, August 05, 2011 7:38 PM

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Perhaps this library will help you:

http://www.oscilloscope-lib.com/

Chris

- Tuesday, August 30, 2011 10:34 PM

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Does this code works in visual studio c# if the incoming data is coming at 256Hz?

I am having trouble displaying multiple line graphs in one panel, the data points seems too much for the program to display and it lags and my laptop runs real loud.

Any other way to draw line graphs in efficient matter (that would not lag when drawing 9 lines in 256Hz)?

Thanks a lot and hope someone has the answer.